

Compiled information on the Lucent RFTG, RFTG-m, RFG-M, RFTG-u, RFTGm-II

Version V2 (29 June 2026)

By Rudi's Electronics Lab

For more recent versions check:

https://github.com/RudisElectronicsLab/Lucent_RFTG_family_documentation



RFTG



(~1996)

RFTG-m



(~1998)

RFG-M



(~2000)

No GPS,
1PPS or
RS422

RFTG-u



(~2000)

No Rb

RFTGm-II



(~2001)

Members of the RFTG / RFG family
(without combiner/splitter)

RFTG



RFTG-m



Must exist; cannot find any picture

RFG-M



RFTG-u



Has no combiner/splitter

RFTGm-II



Members of the RFTG / RFG family

The nomenclature follows the nameplates on the devices.

Model	RFTG		RFTG-m		RFG-M (without the 'T') No GPS		RFTG-u Only as dual XO, no Rb		RFTGm-II	
Introduction year (approx.)	1996, three versions		1998		2000		2000		2000-2001	
Oscillator units (part # if known)	Rb (=REF0): KS24019 L102/L102A XO (=REF1): KS24019 L104/ L104A/L104B		RFTG-m-RB (1) KS24019 L105A RFTG-m-XO KS24019 L106A, L106B, L106C (also seen 2xRb units)		RFG-M-RB WP-92065 L11 RFG-M-XO WP-92066 L10		RFTG-u REF0 Z3812A RFTG-u REF1 Z3811A Pair + harness: Z3810AS		RFTGm-II-Rb KS-24019 L108 RFTGm-II-XO KS-24019 L109	
	Rb (left)	XO (right)	Rb	XO	Rb	XO	XO (REF0)	XO (REF1)	Rb	XO
15 MHz out (sine)	Only via combiner		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
10 MHz out (square)	Yes		Yes		Yes (2x)		Yes		Yes (2x)	
10 MHz Ref In				Yes		Yes				Yes
1PPS (individual units)	No	No	Yes	Yes	No (?)	No (?)	Yes	Yes	Yes	Yes
1PPS (combined)	Yes		Yes		No		No		Yes	
Rb type	?		EFRATOM LPRO-101 (1)		?		No Rb		Symmetricom/Datum LPRO (3)	
OCXO type	?		EFRATOM 105243-001 (1)		?		Milliren (MTI) 260-0624-C (5.000MHz) (2)		EFRATOM 105243-003 (3)	
GPS receiver	In unit		In XO module (can discipline both) (Motorola Oncore)		No GPS		In REF1 module [likely Motorola Oncore]		In XO module (can discipline both) [likely Motorola Oncore]	
Meets Lucent spec.	KS24019		KS24019		?		KS24361		KS24019	
Physical controls	Manual switch for active unit		n/a		n/a		Manual switch for output level		n/a	
Notes	Purpose: Series II AMPS and CDMA system						"Built by HP/ Symmetricom" (16)			

Members of the RFTG family

Model	RFTG	RFTG-m or RFTGm-II	RFG-M (without the 'T')	RFTG-u
Fill ports & wire harness		Rudi's Electronics Lab [36]		
Power port	DSUB-15 (!)	P1', DSUB-9 male +24VDC, Pinout (26) (36)	P1', DSUB-9 male +24VDC	'P1', DSUB-9 male. +24VDC Pinout (14) ; between 18 and 36 Volts (15)
ALARM port	No such port	'J3', DSUB-9 female. Pinout and description, Confirmation –m and m-II are the same (26) [36]	'J3', DSUB-9 female. [No pinout found]	'J3', DSUB-9 female Opto isolator outputs. Pinouts (24) , meaning unknown
INTERFACE port (inter-unit)	[Not ports in the front, but via the frame, somehow the units must be connected]	'J5' DSUB-9 female; Description (9) Cable wiring and pinout (4) (12) ; GPS message routing (12) ; Cable wiring (5) , pinout likely same as (4) ; Full pinout, confirms it is same as –m (8) (26) [36]	DSUB-9 female [No pinout found]	'J5' DSUB-15 female (three-row); Pinout and measurement (13) ; Pinout (incomplete) (14) ; See also (2) ; Pinout & extensive description (15) ; Also has serial port, messages analysed by (14) ; Pinout and description (25) supplemented by (24) , Extensive description (27)
RS422/1PPS port	No separate port per unit, but RS422 and 1PPS are available via the combined 15-pin sockets	'J6' DSUB-9 female Pinout (1) (36) Confirmation –m and m-II are the same (26)	No such port	'J6' DSUB-9 female Pinout & extensive description (15) ; Details of RS422 talker messages (24) ; "speak only when spoken to" (25)
DIAGNOSTIC port	No such port	No such port	No such port	'J8' DSUB-9 female. Pinout and description, SCPI, 'same as J6 without PPS' (15) For commands and monitor status (source)
Connector panel / frame	In between the two units	-m: [Have not seen it, but probably exists?] m-II: yes, wide frame	Wide frame	[Don't think this exists]
Signal combiner/splitter (for 15 MHz)	6 outputs (TNC)	m-II: 2 inputs, 6 outputs (TNC)	2 inputs, 6 outputs (TNC)	No signal combiner/splitter
Signal combiner (for RS422 and 1PPS)	J7, J8, J9 (on frame, each DSUB-15). Each has 1 PPS, alarm and status signals. Uses cable with two DB9 RS422 connectors. (11)	J7, J8, J9 (on frame, each DSUB-15). Each has 1 PPS, alarm and status signals. Cabling (6, 7) [36]	No such port,	No signal combiner/splitter
Schematics	[Not found]	[Not found]	Reverse-engineered by Bud (1, 2)	[Not found]
Publicly available documentation	Reference Frequency and Timing Generator User Documentation Issue 1.0 November 1, 1996 (11) ; Lucent AUTOPLEX Cellular Telecommunications Systems System 1000 (29)	Short note (26)	[Not found]	[Not found]

Members of the RFTG family

Model	RFTG	RFTG-m RFTGm-II	RFG-M (without the 'T')	RFTG-u
RFTG 'Talker' mode with unit status, alarms, GPS time. Works via simple serial monitor (e.g. PuTTY on Windows or SerialTools by Kok Chen W7AY on MacOS).	On J7, J8, J9 (for both Rb and XO)	On Rb J6 and XO J6 Also on J7, J8, J9 on frame (for both Rb and XO)	No (lacks RS422 bus)	On REF0 J6 and REF1 J6. Serial monitor (14)
GPS 'Talker' mode with Motorola Oncore messages every ~5 seconds. @@ Ea (position), @@En (time), @@Bb (visible satellites) and @@Bo (UTC offset)	[Don't think so as there is no interface bus]	Yes, on XO J5 pin 2 (of 9 pins)	No (lacks GPS as well as RS422 bus)	Yes, on REF1 J5 pin1 (of 15 pins)
Command set for interactive control and communication (voltages, GPS data, etc.)	[Proprietary binary command set]	[Proprietary binary command set]	No (lacks RS422 bus)	'J8' DIAGNOSTIC port: Supports SCPI-like HP command set (16)
Software	HP RFTG utility ("RFTG Diagnostic Software package" (use 27 , not 28). (Use Windows XP or older OS). Communicates with both Rb and XO at the same time, when using two serial interfaces.	HP RFTG utility ("RFTG Diagnostic Software package" (use 27 , not 28). (Use Windows XP or older OS). Communicates with both Rb and XO at the same time, when using two serial interfaces. Mark S. Sims Lady Heather v6 Beta (20). Can run simultaneous instances for Rb and XO. Use the J6 RS422/1PPS port Works on Windows XP and Windows 11. Does not speak the "HP SCPI command set" so does not work with Symmetricom's SATSTAT software or the John MacConnell GPS.EXE utility (which looks for :ptim:tcd:cont 0)	No (lacks RS422 bus)	Mark S. Sims Lady Heather v5.01 regular edition (18) Symmetricom SATSTAT software (running in Windows XP), from 'J8' DIAGNOSTIC port) (2) (16) (30) HP GPS Control Program (running in Windows XP), from 'J8' DIAGNOSTIC port) (2) (16) (30) GPSCon software under Debian 7 and Wine (2) See also John MacConnell GPS.EXE and GPS-dos2.exe utilities (30)

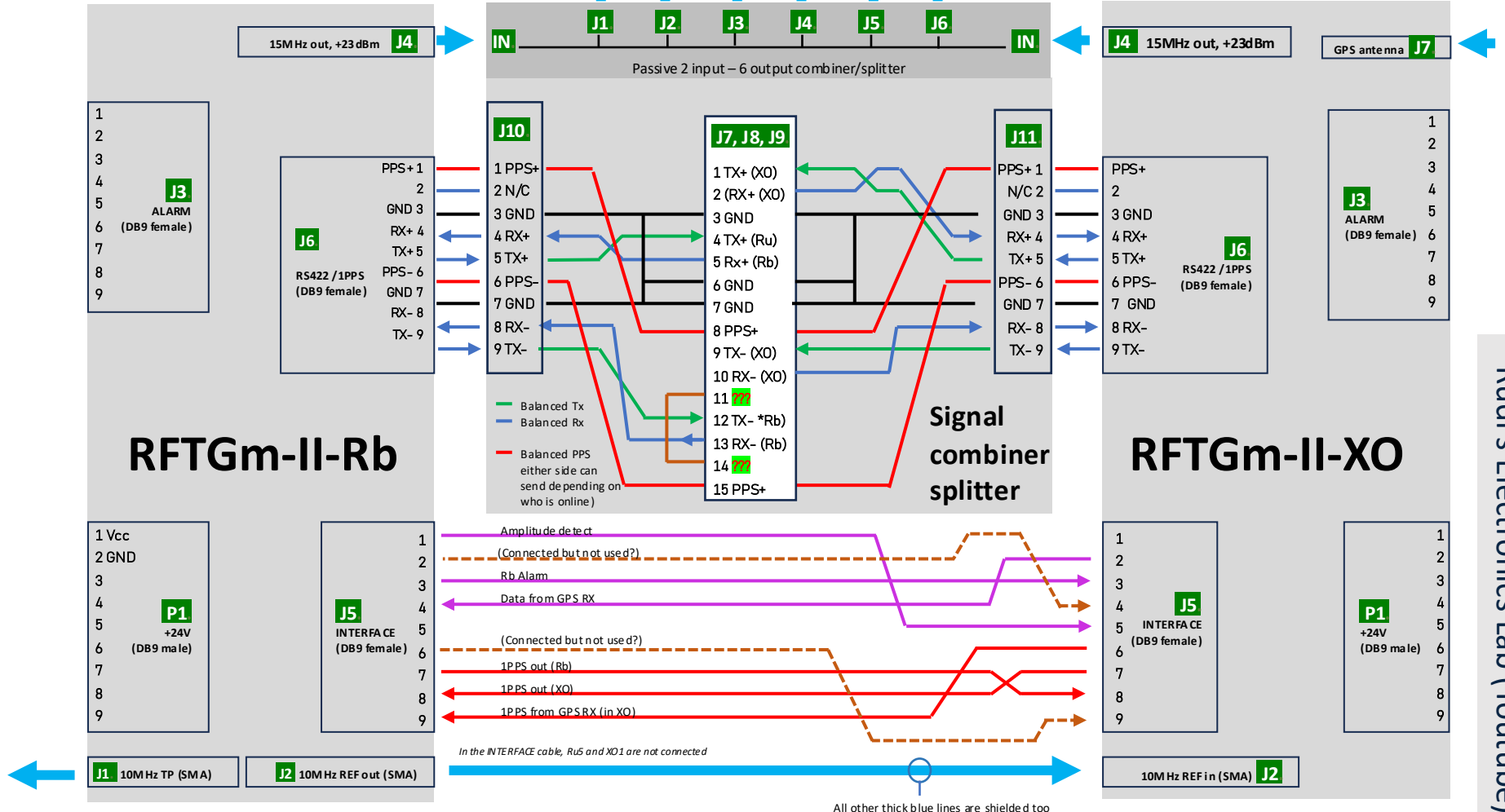
Members of the RFTG family

Model	RFTG	RFTG-m RFTGm-II	RFG-M (without the 'T')	RFTG-u
RFTG 'Talker' mode with unit status, alarms, GPS time. Works via simple serial monitor (e.g. PuTTY on Windows or SerialTools by Kok Chen W7AY on MacOS).	On J7, J8, J9 (for both Rb and XO)	On Rb J6 and XO J6 Also on J7, J8, J9 on frame (for both Rb and XO)	No (lacks RS422 bus)	On REF0 J6 and REF1 J6. Serial monitor (14)
GPS 'Talker' mode with Motorola Oncore messages every ~5 seconds. @@ Ea (position), @@En (time), @@Bb (visible satellites) and @@Bo (UTC offset)	[Don't think so as there is no interface bus]	Yes, on XO J5 pin 2 (of 9 pins)	No (lacks GPS as well as RS422 bus)	Yes, on REF1 J5 pin1 (of 15 pins)
Command set for interactive control and communication (voltages, GPS data, etc.)	[Proprietary binary command set]	[Proprietary binary command set]	No (lacks RS422 bus)	'J8' DIAGNOSTIC port: Supports SCPI-like HP command set (16)
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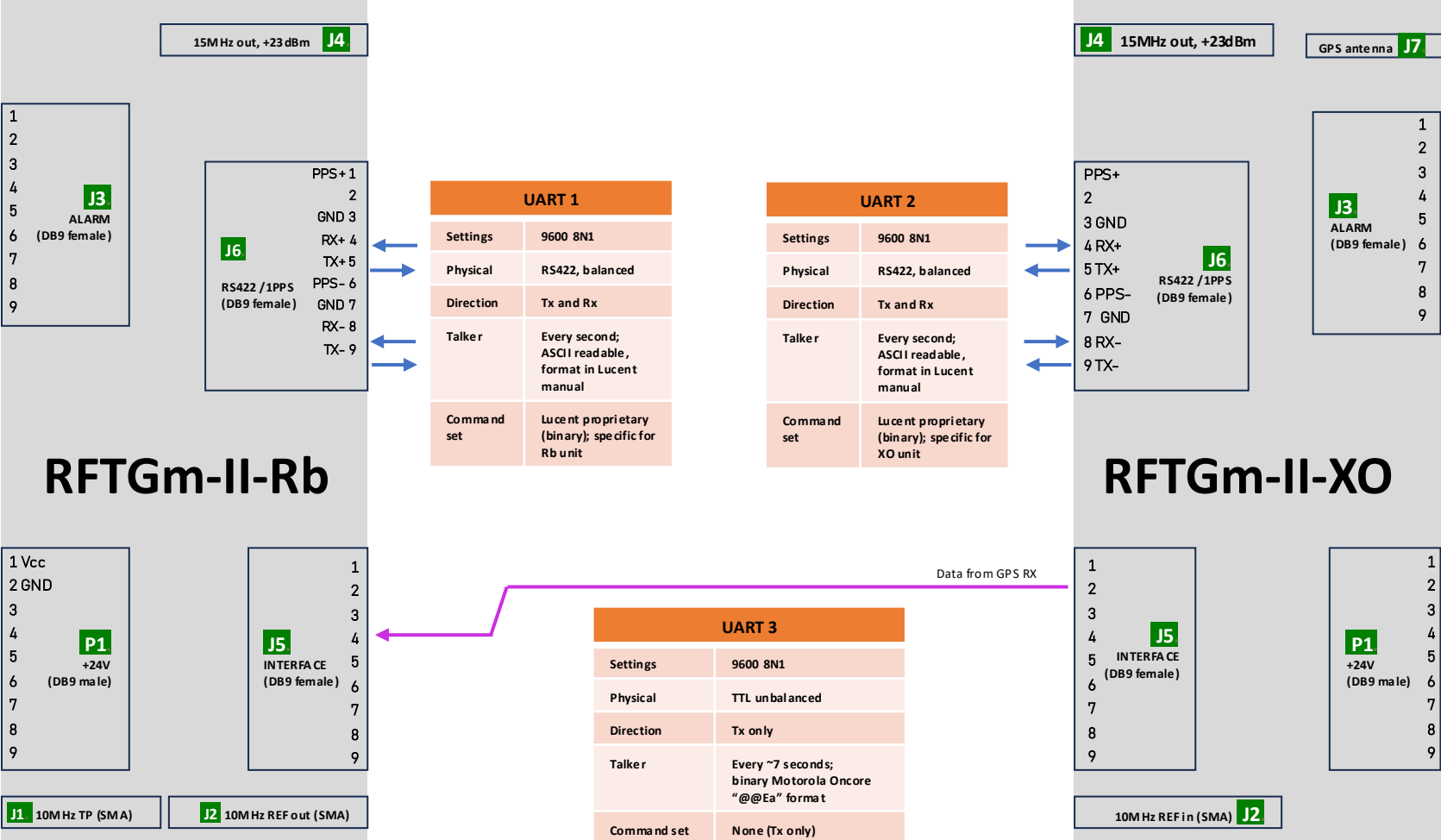
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- (4) Pozdrawiam Wojtek SP7WK <http://bee.mif.pg.gda.pl/ciasteczki/kowypotwor/Lucent/RFTGm%20Interface%20Cable.pdf>
- (5) Rudi's Electronics Lab (via time-nuts). https://febo.com/pipermail/time-nuts_lists.febo.com/2026-May/110668.html
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- (13) Brooke Clarke <https://www.prc68.com/l/KS-24361.html>
- (14) Dan Watson (Sync Channel Blog) <https://syncchannel.blogspot.com/2015/08/standalone-operation-of-lucent-ks-24361.html>
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- (21) Tom Knox (via time-nuts). <https://lists.febo.com/empathy/thread/U62R2ZJPU36CRJSYSRJ5HU4Z6VAUFLT>
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- (35) Peter Garde. <https://www.dropbox.com/sc/fi/6spqc2vbydk7c7jzh6fo/Stand-alone-Lucent-REF-0.pdf?rlkey=sokvbr1koxhtidmmcnib1s68&e=2>
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RFTGm-II connections (1/3)

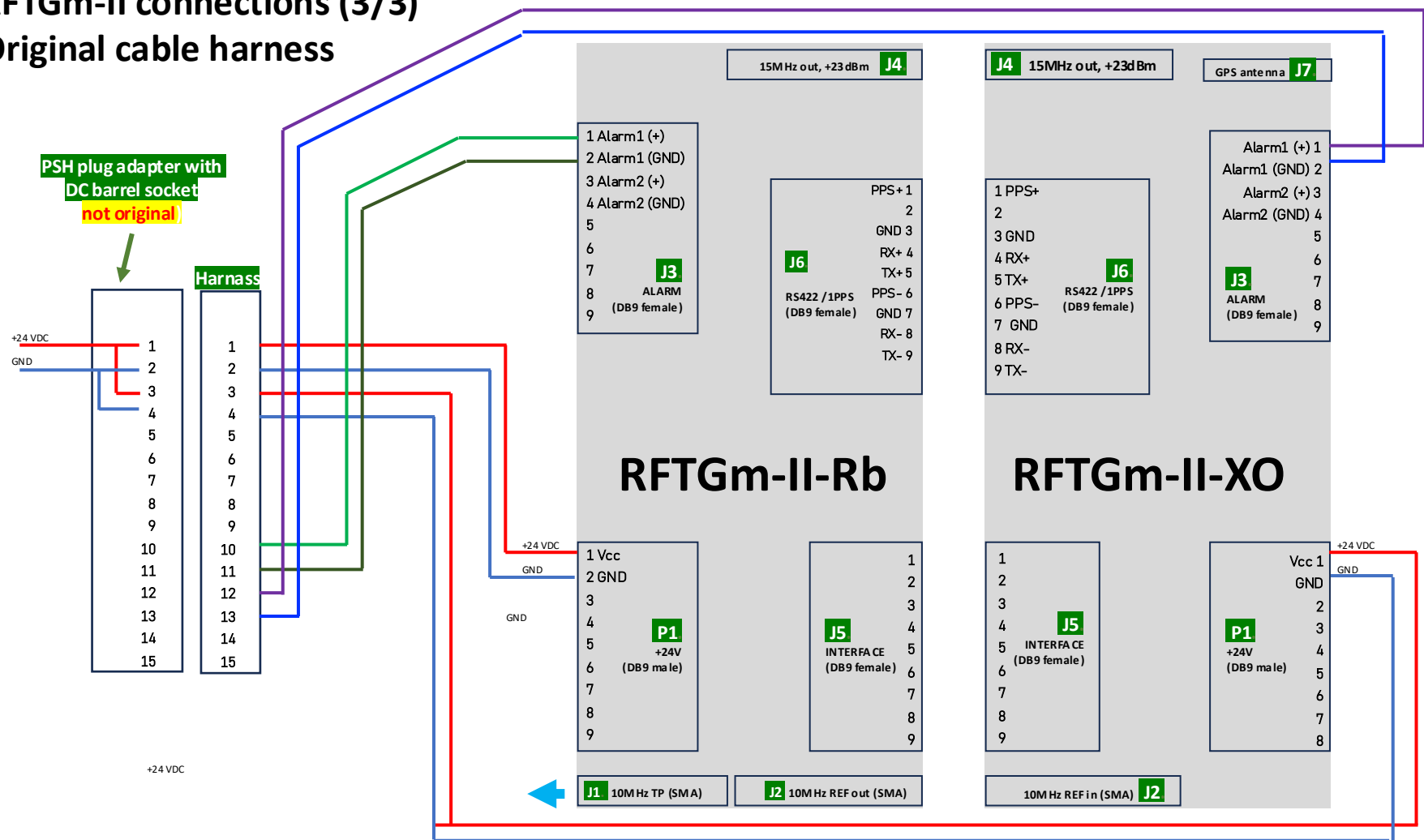


RFTGm-II connections (2/3) Serial protocols



RFTGm-II connections (3/3)

Original cable harness



RFTGm-II-Rb

Default secondary
frequency source

Executes

Puppet

Muscle

RFTGm-II-XO

Backup secondary
frequency source

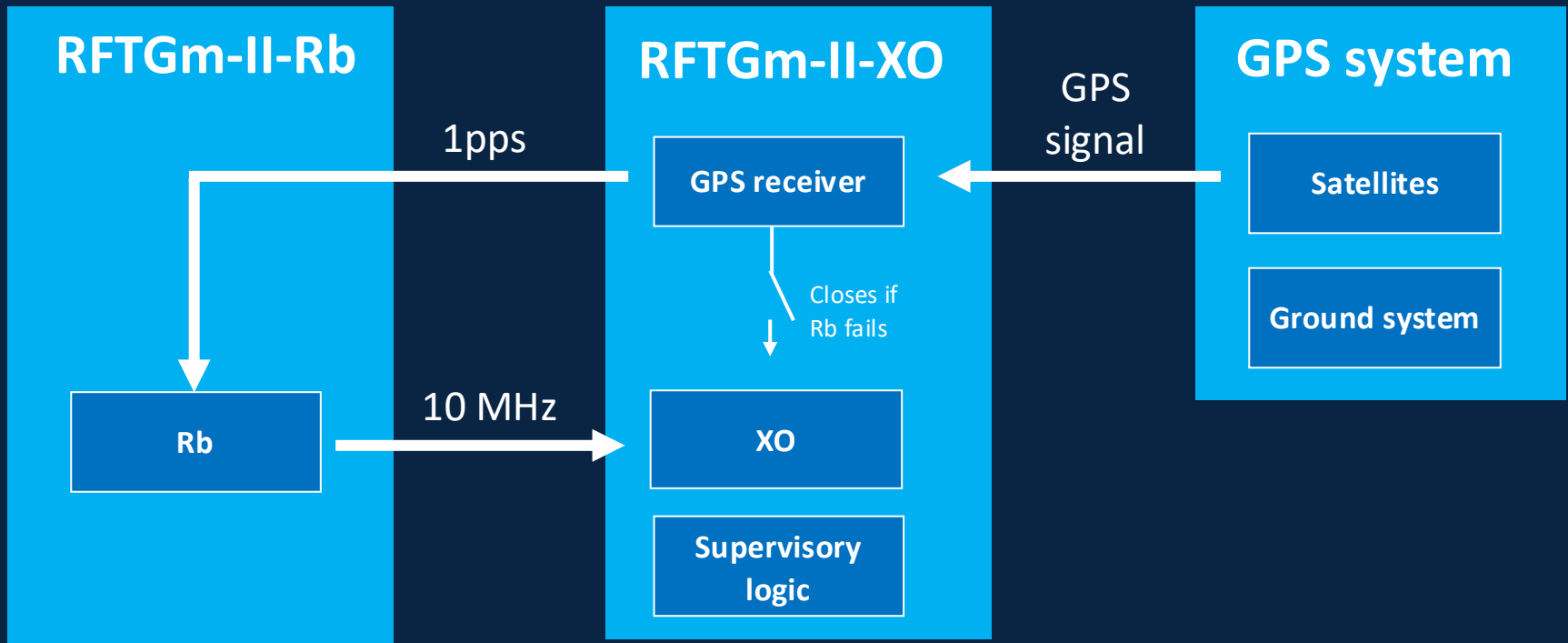
Supervises,
monitors, controls

Master

Brain

GPS system

Primary frequency
source



- The GPS receiver in the XO module does not discipline the XP
- Instead, it sends its 1PPS signal to the Rb unit
- The XO follows the RB using the 10MHz interlink, unless the RB fails
- In case the Rb fails, there is a seamless failover in output of the combiner/splitter
- If Rb recovers, XO very slowly switches back to Rb lock

Formatting of the talker messages on the RS422/1PPS interface

Category	Char position	Meaning				
Preamble	0	Colon character (:)				
Status	1	[0]=Message from Ref 0 [1]=Message from Ref 1				
	2	[0]=Module is active [1]=Module is inactive				
	3	Always set to zero				
	4	[0]=Module is in warm-up mode [1]=Module is locked to GPS [2]=Module is Flywheeling [3]=Module has failed	Bit 0 =1	Bit 1 =1	Bit 2=1	Bit 3 =1
Alarm message	5	Alarm character 1	PPS output missing (*)	1 PPS greater than high threshold (*)	Antenna feed fault	Warm-up failed
	6	Alarm character 2	Oscillator unit maintenance required	GPS discipline board ROM test failed (*)	GPS discipline board RAM failed (*)	15 MHz output failed (*)
	7	Alarm character 3	Flywheeling failed (exceeded 8 hours)	Internal threshold temperature exceeded	1 PPS correction amount between 2 and 9µs	Delta temperature greater than high threshold
	8	Alarm character 4	GPS receiver failed	Oscillator lock failed (*)	Reserved	Oscillator PLL lock failed (*)
	9-10	Total time elapsed since start of operating mode described by the status code position 4				
Other	11-18	GPS Time				
	19-22	Manufacturer/Version				
	23-24	Checksum				
Postamble	25	Carriage Return character				
	26	Line Feed character				

Lucent RFTG utility (“RFTG Diagnostic Software package”)

For RFTG, RFTG-m and RFTGm-II, I believe.

1. Seems designed for Windows 95 (11), but I ran it on Windows XT. Higher versions do not seem to work.
2. Set your screen resolution low (800 x 600, preferably) for RFTG.EXE to render well
3. Get a working RS422 to RS232 converter and an RS232 to USB converter
4. Install chipset drivers for USB converter (like FTDI chipset)
5. In Device Monitor, check whether the new virtual COM device is present, the number assigned (e.g. COM4) and set it to 9600 8N 1
6. (Note that RFTG only supports COM1 to COM4! If assigned to a higher number, try to change that in Device Manager, advanced settings of the COM port)
7. Connect to RS422/1PPS (J6) on either unit
8. Or do this all twice and connect to RS422/1PPS (J6) on both unit
9. Test whether a serial monitor such as PuTTY shows the talking mode messages once a second (set it to the found COM number)
10. Go to <http://www.ko4bb.com/> and search for “GPS Timing RFTG-m Software” (deep link: [27](#)).
11. Download all six files (1DIAG.DAT, GT200.DLL, QCBASED.DLL, QCRTD.DLL, RFTG.EXE, RFTGM.DAT)
12. Note that
13. Place them in a directory (I used c:\Program Files\ko4bb-RFTG)
14. Execute RFTG.EXE (can be done from File Explorer, no need to use terminal)
15. Set the comm ports for Rb resp. XO in MODE | Set Comm Port
16. Select MODE | Normal
17. Consider the manual at (11), which seems to be for the -m version indeed (Screenprints show “RFTGM” and §4.4 talks of ports J7, J8 and J9).
18. If the virtual COM ports are not recognised, even if PuTTY sees them, just keep on trying. Usually takes me several times

PS In my case, many menus (for both Rb and XO) do not work ☹️. Might be a RFTG-m versus RFTGm-II compatibility issue?

PS1. The software from KO4BB is slightly different from the one in the manual (11) (e.g., the table on page 18 top shows items not in the software, and the other way around). For neither can I find a version number

PS2. do NOT use the software from Pozdrawiam Wojtek SP7WK (28). It is only the RFTG.EXE and misses the necessary DLLs!

Lady Heather Beta 6.x on Windows

For RFTG-m and RFTGm-II (the RFTG-u works with Lady Heather 5.0 regular release)

1. Works on many Windows versions (I used both XP and 11)
2. Get a working RS422 to RS232 converter and RS232 to USB converter
3. (Or directly a RS422 to USB converter)
4. Install chip set drivers for USB converter (like FTDI chipset)
5. In Device Monitor, check whether the new virtual COM device is present, what number is has been assigned (e.g. COM4) and set it to 9600 8N1
6. Connect to RS422/1PPS (J6) on either unit
7. Test whether a serial monitor such as PuTTY shows the talking mode messages once a second (set it to the found COM number)
8. Install Lady Heather V5.01 from ([18](#))
9. Copy the entire program directory (present Program Files of Program Files x86) into a new easy-to-reach place like c:\heather6
10. Get the V6 version HEATHER.EXE from ([20](#)). and overwrite the same-named file in c:\heather
11. Start a terminal session (command.com), go to c:\heather6
12. Start Heather up from the terminal window, typing heather /4 (where the number refers to the identified COM port)
13. Run multiple instances of Heather as desired